```
(base) mmcoe@mmcoe-System-Product-Name:~$ sudo mysql
[sudo] password for mmcoe:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 8.0.42-0ubuntu0.20.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> use Pass_4;
Database changed
mysql> CREATE TABLE Customers (
    -> ID INT PRIMARY KEY,
    -> name VARCHAR(15),
    -> age INT,
    -> address VARCHAR(15),
    -> salary INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Orders (
    -> O_Id INT,
    -> O_date DATE,
    -> Cust_Id INT,
    -> amount INT
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> SHOW TABLES;
+------------------+
| Tables_in_Pass_4 |
+------------------+
| Customers        |
| Orders           |
+------------------+
2 rows in set (0.01 sec)
```

```
mysql> DESCRIBE Customers;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| ID      | int         | NO   | PRI | NULL    |       |
| name    | varchar(15) | YES  |     | NULL    |       |
| age     | int         | YES  |     | NULL    |       |
| address | varchar(15) | YES  |     | NULL    |       |
| salary  | int         | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> DESCRIBE Orders;
+---------+------+------+-----+---------+-------+
| Field   | Type | Null | Key | Default | Extra |
+---------+------+------+-----+---------+-------+
| O_Id    | int  | YES  |     | NULL    |       |
| O_date  | date | YES  |     | NULL    |       |
| Cust_Id | int  | YES  |     | NULL    |       |
| amount  | int  | YES  |     | NULL    |       |
+---------+------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> SELECT * FROM Customers;
Empty set (0.00 sec)

mysql> SELECT * FROM Orders;
Empty set (0.00 sec)

mysql> INSERT INTO Customers (ID, name, age, address, salary) VALUES
    -> (1, 'Alice', 25, 'New York', 22000),
    -> (2, 'Bob', 30, 'Los Angeles', 18000),
    -> (3, 'Charlie', 22, 'Chicago', 25000),
    -> (4, 'David', 28, 'Houston', 21000),
    -> (5, 'Eve', 24, 'Miami', 19000);
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> INSERT INTO Orders (O_Id, O_date, Cust_Id, amount) VALUES
    -> (101, '2025-01-15', 1, 500),
    -> (102, '2025-02-10', 2, 750),
    -> (103, '2025-03-20', 1, 300),
    -> (104, '2025-04-05', 4, 1200),
    -> (105, '2025-05-12', 5, 450);
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Customers;
+----+---------+------+-------------+--------+
| ID | name    | age  | address     | salary |
+----+---------+------+-------------+--------+
|  1 | Alice   |   25 | New York    |  22000 |
|  2 | Bob     |   30 | Los Angeles |  18000 |
|  3 | Charlie |   22 | Chicago     |  25000 |
|  4 | David   |   28 | Houston     |  21000 |
|  5 | Eve     |   24 | Miami       |  19000 |
+----+---------+------+-------------+--------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Orders;
+------+------------+---------+--------+
| O_Id | O_date     | Cust_Id | amount |
+------+------------+---------+--------+
|  101 | 2025-01-15 |       1 |    500 |
|  102 | 2025-02-10 |       2 |    750 |
|  103 | 2025-03-20 |       1 |    300 |
|  104 | 2025-04-05 |       4 |   1200 |
|  105 | 2025-05-12 |       5 |    450 |
+------+------------+---------+--------+
5 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Orders (O_Id, O_date, Cust_Id, amount) VALUES
    -> (101, '2025-01-15', 1, 500),
    -> (102, '2025-02-10', 2, 750),
    -> (103, '2025-03-20', 1, 300),
    -> (104, '2025-04-05', 4, 1200),
    -> (105, '2025-05-12', 5, 450);
Query OK, 5 rows affected (0.00 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Customers;
+----+---------+------+-------------+--------+
| ID | name    | age  | address     | salary |
+----+---------+------+-------------+--------+
|  1 | Alice   |   25 | New York    |  22000 |
|  2 | Bob     |   30 | Los Angeles |  18000 |
|  3 | Charlie |   22 | Chicago     |  25000 |
|  4 | David   |   28 | Houston     |  21000 |
|  5 | Eve     |   24 | Miami       |  19000 |
+----+---------+------+-------------+--------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Orders;
+------+------------+---------+--------+
| O_Id | O_date     | Cust_Id | amount |
+------+------------+---------+--------+
|  101 | 2025-01-15 |       1 |    500 |
|  102 | 2025-02-10 |       2 |    750 |
|  103 | 2025-03-20 |       1 |    300 |
|  104 | 2025-04-05 |       4 |   1200 |
|  105 | 2025-05-12 |       5 |    450 |
+------+------------+---------+--------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT c.ID, c.name, o.amount, o.O_date
    -> FROM Customers c
    -> LEFT JOIN Orders o
    -> ON c.ID = o.Cust_Id;
+----+---------+--------+------------+
| ID | name    | amount | O_date     |
+----+---------+--------+------------+
|  1 | Alice   |    300 | 2025-03-20 |
|  1 | Alice   |    500 | 2025-01-15 |
|  2 | Bob     |    750 | 2025-02-10 |
|  3 | Charlie |   NULL | NULL       |
|  4 | David   |   1200 | 2025-04-05 |
|  5 | Eve     |    450 | 2025-05-12 |
+----+---------+--------+------------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT c.ID, c.name, o.amount, o.O_date
    -> FROM Customers c
    -> RIGHT JOIN Orders o
    -> ON c.ID = o.Cust_Id;
+------+-------+--------+------------+
| ID   | name  | amount | O_date     |
+------+-------+--------+------------+
|    1 | Alice |    500 | 2025-01-15 |
|    2 | Bob   |    750 | 2025-02-10 |
|    1 | Alice |    300 | 2025-03-20 |
|    4 | David |   1200 | 2025-04-05 |
|    5 | Eve   |    450 | 2025-05-12 |
+------+-------+--------+------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT c.ID, c.name, o.amount, o.O_date
    -> FROM Customers c
    -> LEFT JOIN Orders o ON c.ID = o.Cust_Id
    -> UNION ALL
    -> SELECT c.ID, c.name, o.amount, o.O_date
    -> FROM Customers c
    -> RIGHT JOIN Orders o ON c.ID = o.Cust_Id;
+------+---------+--------+------------+
| ID   | name    | amount | O_date     |
+------+---------+--------+------------+
|    1 | Alice   |    300 | 2025-03-20 |
|    1 | Alice   |    500 | 2025-01-15 |
|    2 | Bob     |    750 | 2025-02-10 |
|    3 | Charlie |   NULL | NULL       |
|    4 | David   |   1200 | 2025-04-05 |
|    5 | Eve     |    450 | 2025-05-12 |
|    1 | Alice   |    500 | 2025-01-15 |
|    2 | Bob     |    750 | 2025-02-10 |
|    1 | Alice   |    300 | 2025-03-20 |
|    4 | David   |   1200 | 2025-04-05 |
|    5 | Eve     |    450 | 2025-05-12 |
+------+---------+--------+------------+
11 rows in set (0.04 sec)

mysql> SELECT A.name, B.name, A.address
    -> FROM Customers A, Customers B
    -> WHERE A.ID <> B.ID AND A.address = B.address;
Empty set (0.03 sec)

mysql> SELECT A.name, B.name, A.address
    -> FROM Customers A, Customers B
    -> WHERE A.address = B.address;
+---------+---------+-------------+
| name    | name    | address     |
+---------+---------+-------------+
| Alice   | Alice   | New York    |
| Bob     | Bob     | Los Angeles |
| Charlie | Charlie | Chicago     |
| David   | David   | Houston     |
| Eve     | Eve     | Miami       |
+---------+---------+-------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT c.ID, c.name, o.amount, o.O_date
    -> FROM Customers c
    -> CROSS JOIN Orders o;
+----+---------+--------+------------+
| ID | name    | amount | O_date     |
+----+---------+--------+------------+
|  5 | Eve     |    500 | 2025-01-15 |
|  4 | David   |    500 | 2025-01-15 |
|  3 | Charlie |    500 | 2025-01-15 |
|  2 | Bob     |    500 | 2025-01-15 |
|  1 | Alice   |    500 | 2025-01-15 |
|  5 | Eve     |    750 | 2025-02-10 |
|  4 | David   |    750 | 2025-02-10 |
|  3 | Charlie |    750 | 2025-02-10 |
|  2 | Bob     |    750 | 2025-02-10 |
|  1 | Alice   |    750 | 2025-02-10 |
|  5 | Eve     |    300 | 2025-03-20 |
|  4 | David   |    300 | 2025-03-20 |
|  3 | Charlie |    300 | 2025-03-20 |
|  2 | Bob     |    300 | 2025-03-20 |
|  1 | Alice   |    300 | 2025-03-20 |
|  5 | Eve     |   1200 | 2025-04-05 |
|  4 | David   |   1200 | 2025-04-05 |
|  3 | Charlie |   1200 | 2025-04-05 |
|  2 | Bob     |   1200 | 2025-04-05 |
|  1 | Alice   |   1200 | 2025-04-05 |
|  5 | Eve     |    450 | 2025-05-12 |
|  4 | David   |    450 | 2025-05-12 |
|  3 | Charlie |    450 | 2025-05-12 |
|  2 | Bob     |    450 | 2025-05-12 |
|  1 | Alice   |    450 | 2025-05-12 |
+----+---------+--------+------------+
25 rows in set (0.00 sec)
```

```
mysql> SELECT *
    -> FROM Customers
    -> WHERE ID IN (
    ->     SELECT ID
    ->     FROM Customers
    ->     WHERE salary > 20000
    -> );
+----+---------+------+----------+--------+
| ID | name    | age  | address  | salary |
+----+---------+------+----------+--------+
|  1 | Alice   |   25 | New York |  22000 |
|  3 | Charlie |   22 | Chicago  |  25000 |
|  4 | David   |   28 | Houston  |  21000 |
+----+---------+------+----------+--------+
3 rows in set (0.03 sec)
```

```
mysql> CREATE TABLE cust_bkp AS
    -> SELECT *
    -> FROM Customers;
Query OK, 5 rows affected (0.11 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql> DESCRIBE cust_bkp;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| ID      | int         | NO   |     | NULL    |       |
| name    | varchar(15) | YES  |     | NULL    |       |
| age     | int         | YES  |     | NULL    |       |
| address | varchar(15) | YES  |     | NULL    |       |
| salary  | int         | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM cust_bkp;
+----+---------+------+-------------+--------+
| ID | name    | age  | address     | salary |
+----+---------+------+-------------+--------+
|  1 | Alice   |   25 | New York    |  22000 |
|  2 | Bob     |   30 | Los Angeles |  18000 |
|  3 | Charlie |   22 | Chicago     |  25000 |
|  4 | David   |   28 | Houston     |  21000 |
|  5 | Eve     |   24 | Miami       |  19000 |
+----+---------+------+-------------+--------+
5 rows in set (0.03 sec)

mysql> UPDATE Customers
    -> SET salary = salary * 1.1
    -> WHERE age >= 24
    -> AND ID IN (
    ->     SELECT ID FROM cust_bkp
    -> );
Query OK, 4 rows affected (0.04 sec)
Rows matched: 4  Changed: 4  Warnings: 0

mysql> DELETE FROM Customers
    -> WHERE ID IN (
    ->     SELECT ID
    ->     FROM cust_bkp
    ->     WHERE age > 26
    -> );
Query OK, 2 rows affected (0.04 sec)

mysql> SELECT * FROM Customers;
+----+---------+------+----------+--------+
| ID | name    | age  | address  | salary |
+----+---------+------+----------+--------+
|  1 | Alice   |   25 | New York |  24200 |
|  3 | Charlie |   22 | Chicago  |  25000 |
|  5 | Eve     |   24 | Miami    |  20900 |
+----+---------+------+----------+--------+
3 rows in set (0.00 sec)
```